AS-Depth: Unleashing OFA Collaboration For Efficient Metric Depth Completion on SoC

Anonymous Author(s)
Anonymous Institution
Anonymous

***Abstract*—Depth completion recovers dense metric depth from the sparse and noisy measurements of low-cost RGB-D sensors, and has become a key enabler for robotic manipulation and autonomous driving. Recent Transformer-based models have achieved promising accuracy on metric depth prediction, but their heavy computation and large parameter counts confine them to run real-time framerate on edge platforms. While modern robotic systems demand real-time depth on resource-constrained edge SoCs. To address this limitation, we observe that this gap lies in the model itself and how the pipeline is scheduled: every frame is pushed through the full network, even though consecutive depth maps at high frame rates are highly redundant. Building on this insight, we present AS-DEPTH, an asynchronous framework to accelerate depth completion on SoCs through three techniques: (1) an asynchronous execution path that decouples depth output from model inference; (2) a heterogeneous collaboration that offloads inter-frame depth propagation to the SoC's on-chip Optical Flow Accelerator (OFA), alongside the GPU running the model; and (3) an adaptive frame-adjustment strategy that tunes the inference rate at runtime. On NVIDIA Jetson AGX Thor, AS-DEPTH delivers over 6× speedup and 50% energy savings over CDM with negligible accuracy drop. Moreover, AS-DEPTH can integrate into existing depth-completion models.**

***Index Terms*—Computer systems organization → Sensors and actuators; Computing methodologies → Perception; Vision for robotics.**

I. INTRODUCTION

Metric depth measures the distance of each pixel in an image from the camera, providing geometry cues such as spatial and shape information that the 2D color image cannot provide. Such depth information can be inputted into the backbone model of applications such as autonomous driving and robotics shown in Fig. 1, which greatly improves the capability of spatial reasoning and enables precision interaction with the objects.

With the widespread availability of commodity depth cameras, acquiring depth information appears to be low cost and straightforward. However, commodity RGB-D cameras frequently suffer from missing values, depth holes, flying pixels, and measurement noise, especially under challenging materials such as transparent, reflective, and specular surfaces. Although recent studies have integrated depth into robotic manipulations, performance remain limited by the poor quality of depth data produced by such device. Therefore, evaluations of robotics is typically restricted to simulators where clean depth is available. Thus, depth completion emerges as an effective method for generating dense metric depth from commodity depth cameras, and has become a critical component for bridging the Sim to Real gap.

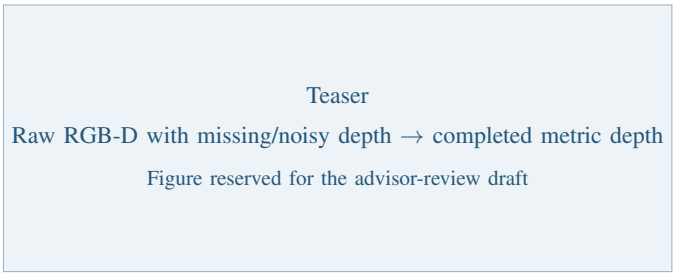


Fig. 1. Teaser. *Pending*

Recent studies 1234 have demonstrated advanced accuracy of depth completion, but their increasingly large Transformer-based architectures impose substantial computation costs. The selected GPU baselines range from 4.26 to 6.62 FPS on RTX 3090/4090 platforms. Because the models and platforms differ, these figures provide context rather than a controlled speedup comparison. Achieving real-time depth completion on an edge SoC such as NVIDIA Jetson AGX Thor remains challenging under its computation budget, motivating algorithm and hardware acceleration.

To address this, we propose As-Depth, a asynchronous heterogeneous acceleration system built on SoC and FPGA tailored for depth completion task on edge platforms, as shown in Fig. 2. As-Depth achieves efficient real time rendering speed and SOTA rendering quality on the edge platform. At the architecure level, it decouples the image representation encoding from the metric depth decoding stage, enabling these two tasks to be executed asynchronously on the SoC and FPGA according to their computational characteristics. On the algorithmn-hardware co-design level, our depth aware quantization and customized decoder design effectively preserve the accuracy of metric depth while reducing computational overhead. At the accelerator level, our depth aware accelerator integrates an ethernet interface unit and a depth decoding engine to handle the decoding pipeline, enabling efficient data communication and computation.

Building on these insights above, we develop a novel heterogeneous acceleration system, As-Depth, that combines the strengths of both the SoC and the FPGA to achieve real-time depth completion on edge platforms. It leverages the mature CUDA cores and Tensor Cores of the SoC to

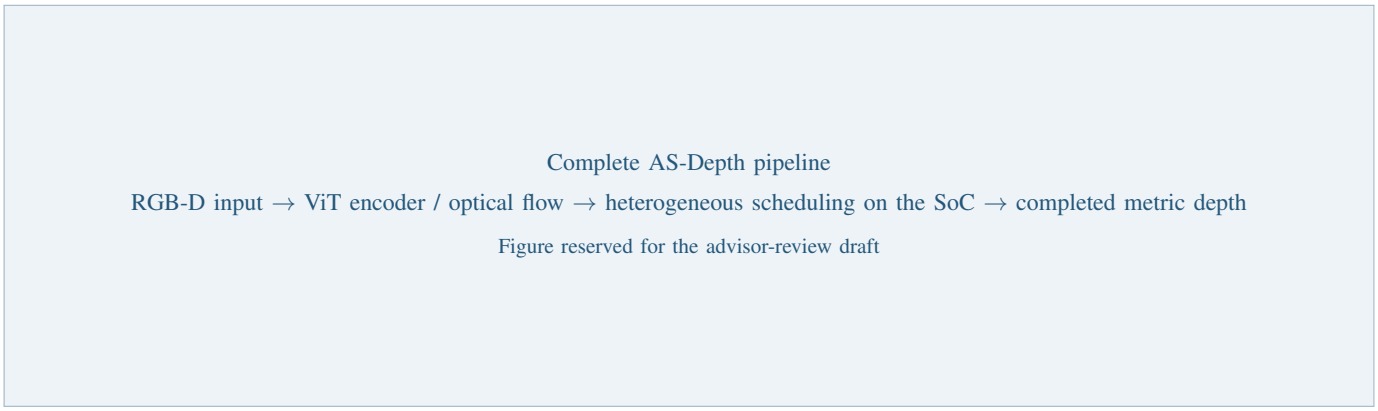


Fig. 2. Overview of the complete AS-DEPTH pipeline. *Pending*

accelerate the transformer-based encoder, while offloading the CNN decoder to our customized FPGA implementation. This design ensures the maximum utilization of both the SoC and the FPGA, enableing pipelined execution to hide the latency of the offloaded operations. Through our quantization aware training and depth aware customized precision, our system strike a balance between performance and resource overhead. Our contributions are as follows:

- We propose a novel heterogenous system and combine task level asynchronous execution for depth completion based on the existing depth completion pipeline, efficiently mapping computing units according to the different computation characteristics of the SoC and FPGA, and using asynchronous optical queued communication to improve inference performance.
- We design a depth-aware quantization strategy, and a customized precision scheme for the fpga implementation. Our approach successfully balance the accuracy of metric depth between model inference latency.
- We implement a prototype of our As-Depth system on the Nvidia Jetson AGX Thor and Intel Stratrix FPGA platform. Our prototype runs at real-time speed of 30FPS, demonstrating the feasibility of our heterogeneous acceleration approach.
- We implement and evaluate As-Depth on Synopsys. Experiments show that our system achieves 5.4x speedup and xxx improvement in energy efficiency over the latest SOTA implementation on the targeted SoC with the whole depth completion pipeline. To the best of our knowledge, this is the first work that achieves real-time 30FPS inference of depth completion on edge platforms. Our system enables real-time depth near sensor possible which contribute to mitigate the Sim to Real gap in the future rbotics applications. Moreover, our approach demonstrates the potential for heterogeneous acceleration in resource-constrained environments, paving the way for future advancements in edge AI systems.

II. RELATED WORKS

A. Depth Completion

Depth completion estimates a dense metric-depth map from an RGB image and an incomplete depth observation [1]. The input depth may contain holes, outliers, and material-dependent noise, whereas the output must retain both absolute scale and local geometry for downstream robotic perception. Modern methods commonly pair a representation encoder with a dense prediction decoder. The encoder extracts global visual and geometric features; the decoder fuses and upsamples these features to the target depth resolution.

B. ViT Compression

Prior GPU depth-completion systems establish the accuracy of large models, and prior FPGA accelerators demonstrate the efficiency of specialized low-precision datapaths. AS-DEPTH connects these directions by retaining the encoder on the SoC, specializing the decoder on FPGA, and treating the model boundary and cross-frame schedule as first-class parts of the design.

Draft note: General ViT compression techniques, e.g., [2]; contrast with AS-DEPTH*, whose optimizations are application specific to depth completion.*

III. METHOD

AS-DEPTH combines three mechanisms derived from the observations below. A task-level execution framework partitions the model and overlaps consecutive frames; DAQ coordinates low precision across the Thor encoder and FPGA decoder; and a dedicated decoder accelerator turns the regular decoder workload into an efficient hardware dataflow.

A. Preliminary

AS-DEPTH preserves the encoder–decoder decomposition of depth completion and uses its boundary as a system partition. The selected depth-completion network and its trained weights are the workload being accelerated, while this paper focuses on the low-precision deployment, heterogeneous execution, and dedicated decoder hardware.

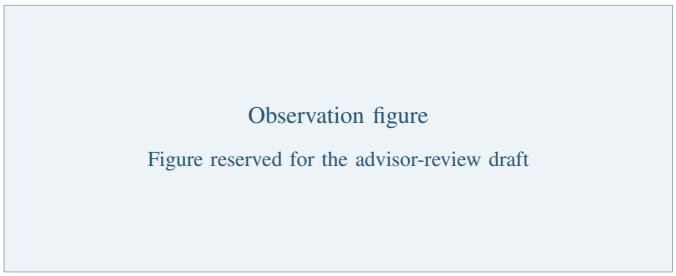


Fig. 3. Observation. *Pending*

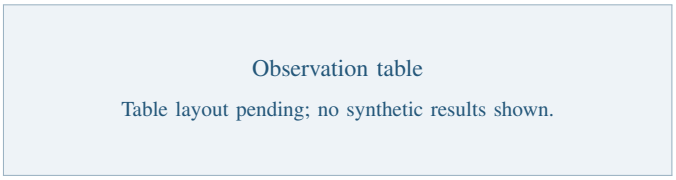
a) Computational characteristics: The two stages expose different opportunities for edge acceleration. The encoder contains Transformer blocks whose matrix operations and MLP layers can use Thor's mature GPU and Tensor Core execution stack. The decoder is composed primarily of structured dense-prediction operators for which customized parallelism, buffering, and precision can be implemented on FPGA. This difference motivates a stage-level mapping rather than repeatedly moving individual operators between devices. We profile the unmodified model on Thor and report the latency of preprocessing, encoder, decoder, and postprocessing under a fixed resolution, precision, batch size, and power mode (Fig. 3 and Table I).

b) Edge SoC–FPGA heterogeneous execution: An edge SoC offers a complete software environment and high-throughput tensor units, but its compute and energy budget is shared with sensing, planning, and control. An FPGA provides configurable precision, dataflow, and memory organization for a stable subgraph. The combination is useful only when the partition is coarse enough to limit communication and when execution across devices is scheduled to hide, rather than merely add, transfer latency.

c) Observation 1: Monolithic execution misses the real-time budget: The Transformer encoder and CNN decoder have different operator mixes and hardware preferences, yet a conventional deployment executes both stages sequentially on one processor. The resulting end-to-end latency includes the full encoder, decoder, and runtime path for every frame. Merely moving the decoder to another device does not remove this serialization if communication and accelerator execution remain on the critical path. The clear encoder–decoder boundary instead permits a coarse-grained partition: the Thor executes the encoder while the FPGA executes the decoder. Across a stream of frames, queued communication and double buffering can overlap the encoder of frame $i+1$ with transfer and decoding of frame i.

d) Observation 2: End-to-end low precision can degrade metric depth: Low precision is beneficial on both sides of the partition. In particular, selected MLP computation in the Thor-side encoder and the FPGA-side decoder can be executed at reduced precision. A uniform quantization policy, however, must simultaneously accommodate Transformer activation distributions, depth-regression sensitivity, and the numerical contract at the cross-device boundary. Aggressive quantization can therefore save computation and bandwidth while introducing scale errors or local depth artifacts. Rather than treating encoder quantization, feature transfer, and decoder quantization as independent optimizations, AS-DEPTH coordinates their precision choices.

TABLE I
OBSERVATION TABLE. *Pending*



B. Our Design: Optical Flow

Draft note: Optical-flow design, illustrated in Fig. 4.

a) Depth-aware quantization: DAQ treats quantization as an end-to-end property of the heterogeneous model. On Thor, it targets selected MLP computation in the encoder; at the partition boundary, it defines the representation transmitted to the FPGA; and inside the FPGA, it determines the precision of decoder weights, activations, accumulators, and output depth. Training optimizes the quantized path against metric-depth supervision so that numerical error is weighted by its effect on the final depth map rather than by feature reconstruction alone, extending the standard quantization-aware training setting [2].

b) Dedicated depth decoder accelerator: The FPGA accelerator specializes the weighted and auxiliary operators in the depth decoder while retaining a programmable layer schedule. Its datapath must expose the precision contract established by DAQ, reuse weights and activations on chip where profitable, and stream larger intermediate tensors through external memory without changing model semantics. The communication front end converts received feature packets into accelerator-visible buffers and returns the completed depth map after the final layer. This design follows the broader principle of mapping quantized networks to customized FPGA dataflows [3] while specializing the execution path for metric-depth decoding. For every layer, the scheduler records tensor dimensions, addresses, quantization parameters, and dependencies, and verification compares each hardware layer boundary against the quantized software reference.

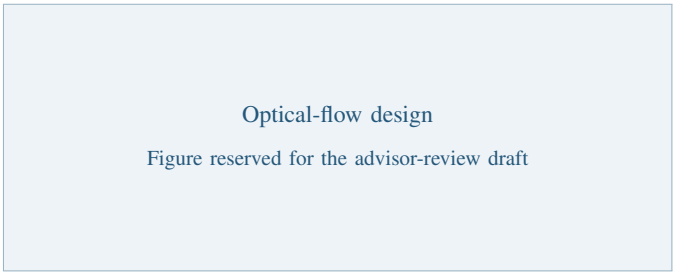


Fig. 4. Optical-flow design. *Pending*

C. Scheduling in Heterogeneous System

a) Model partition and tensor contract: The model is partitioned once at the encoder–decoder boundary. Thor consumes the RGB image and sensor depth, executes the encoder, and exports a quantized feature tensor. The FPGA consumes this tensor and returns the completed metric-depth map.

b) Queued cross-frame execution: The host runtime assigns separate buffers to encoder output, outbound transfer, decoder input, and completed depth. Producer–consumer queues make ownership explicit and prevent a buffer from being reused before its consumer finishes. After pipeline fill, the encoder for the next frame can progress in parallel with transfer and decoder execution for the current frame, as summarized in Algorithm 1.

c) Communication and back pressure: The runtime batches feature metadata with the boundary tensor and preserves frame order on the return path. If either device becomes the bottleneck, bounded queues apply back pressure instead of silently dropping or reordering frames. Section IV measures transfer time separately from useful computation so that communication hiding can be verified.

IV. Experiments

We evaluate AS-Depth by answering four questions. First, can heterogeneous execution sustain the 30-FPS real-time target once communication and runtime overheads are included? Second, does DAQ preserve metric-depth quality on challenging sensor data? Third, how does the complete system compare with recent depth-completion methods? Finally, what resource, communication, power, and energy costs are required to obtain these gains?

A. Settings

a) Baselines: The controlled system comparison runs the same depth-completion network in three deployments: *GPU-only* on an RTX 4090, *Thor-only* on the target edge SoC, and AS-Depth. The GPU-only and Thor-only baselines execute the complete model sequentially on one device; AS-Depth includes feature transfer and output reconstruction in its end-to-end time. For model-level context, we additionally compare against CDM, PriorDA, PromptDA, and OMNI-DC, using their released checkpoints and official evaluation procedures whenever available. Because their published FPS values come from different GPU platforms, we label the execution platform of every method and do not treat cross-platform FPS as a controlled speedup claim.

Algorithm 1 Heterogeneous scheduling (placeholder)

Require: Input frame stream *Pending*
Ensure: Completed metric-depth stream *Pending*
1: **for** each incoming frame **do**
2: *Pending*
3: **end for**

b) Benchmarks: We evaluate all accuracy-oriented experiments on HAMMER and ClearPose. These datasets exercise different object geometries and difficult sensing conditions, so results are reported for each dataset rather than averaged into one score. Every method uses the same test split, validity mask, crop, input resolution, and depth range within a dataset. The system and ablation experiments use batch size one, which matches continuous RGB-D capture on the target robotic platform.

c) Experiment platform: The prototype comprises an NVIDIA Jetson AGX Thor and an Intel Stratix FPGA connected through the system interface described in Section III-C. The FPGA clock, external-memory configuration, link rate, Thor power mode, and software versions are held fixed across all AS-Depth variants. The RTX 4090 baseline uses the same model weights, input resolution, preprocessing, and output validation.

d) Metrics and measurement protocol: We measure both single-frame latency and steady-state throughput. Latency is decomposed into preprocessing, encoder execution, boundary-tensor transfer, decoding, return transfer, and output reconstruction; FPS is measured only after the asynchronous pipeline is filled. Timing includes synchronization needed for a valid completed-depth output, rather than reporting kernel time alone. Depth quality is evaluated with AbsRel, RMSE, MAE, and $\delta_{1.05}$. Lower is better for the three error metrics and higher is better for $\delta_{1.05}$. We report system power, energy per completed frame, and FPS/W under a consistent measurement boundary.

B. Main Experiments

a) Throughput: Table II compares the inference rate of CDM, PriorDA, PromptDA, OMNI-DC, and AS-Depth, together with the execution platform of each method. This comparison answers a deployment question—whether current depth-completion methods can service a real-time sensor stream—while the controlled GPU-only/Thor-only comparison supports architectural speedup claims. We report AS-Depth's measured end-to-end FPS and identify whether its advantage comes from the coarse model partition, cross-frame overlap, or low-precision execution, using the ablation evidence rather than the cross-platform numbers alone.

b) Depth accuracy: Table II also compares all methods under common HAMMER and ClearPose protocols. The discussion reports the absolute metric for every method, AS-Depth's gap to the best-performing method, and its rank on each dataset. The relevant question is whether AS-Depth reaches real-time edge throughput while keeping the accuracy loss small and consistent across both datasets.

C. Ablation Studies

a) Optical flow versus frequency: *Draft note: Ablation on optical flow at different rates, shown in Fig. 5.*

b) Scheduling: *Draft note: Ablation on scheduling, reported in Table III.*

TABLE II
MAIN RESULTS. VALUES MARKED "–" AWAIT MEASUREMENT.

Method	Platform	FPS ↑	HAMMER			ClearPose		
			AbsRel ↓	RMSE ↓	$\delta_{1.05}$ ↑	AbsRel ↓	RMSE ↓	$\delta_{1.05}$ ↑
CDM	–	–	–	–	–	–	–	–
PriorDA	–	–	–	–	–	–	–	–
PromptDA	–	–	–	–	–	–	–	–
OMNI-DC	–	–	–	–	–	–	–	–
AS-DEPTH	–	–	–	–	–	–	–	–

c) Heterogeneous execution: Thor-only versus AS-DEPTH is the primary controlled comparison for the heterogeneous partition: both process the same frames with the same network, while only the latter offloads the decoder and overlaps encoder, communication, and decoder work across frames. We therefore report not only their FPS difference, but also the stage-latency breakdown and the fraction of transfer hidden by the asynchronous schedule. GPU-only is retained as a high-end performance reference, not as an edge-device energy baseline.

d) Metric-depth accuracy: The decisive accuracy comparison is AS-DEPTH without versus with DAQ because their model partition, target bit widths, and runtime are identical. Any quality difference therefore measures the value of coordinating the quantized encoder MLP, boundary representation, and decoder using metric-depth supervision. We report each dataset independently and discuss whether improvements come from reducing large depth errors, increasing the fraction of accurate pixels, or both.

TABLE III
ABLATION ON SCHEDULING. *Pending*

Ablation on scheduling

Table layout pending; no synthetic results shown.

e) Fallback optical flow on CUDA: *Draft note: Optical flow falls back to CUDA; results in Table IV.*

f) Power and energy efficiency: Power is measured for Thor-only and the complete heterogeneous system using the same input stream and steady-state window used for FPS. We derive energy per frame as system power divided by completed frames per second and additionally report FPS/W. The analysis separates the extra power introduced by the additional hardware from the energy saved by shortening the end-to-end execution time.

Optical flow vs. Hz

Measurements pending; no synthetic results shown.

Fig. 5. Ablation on optical flow versus frequency. *Pending*

TABLE IV
FALLBACK OPTICAL FLOW ON CUDA. *Pending*

Fallback optical flow on CUDA

Table layout pending; no synthetic results shown.

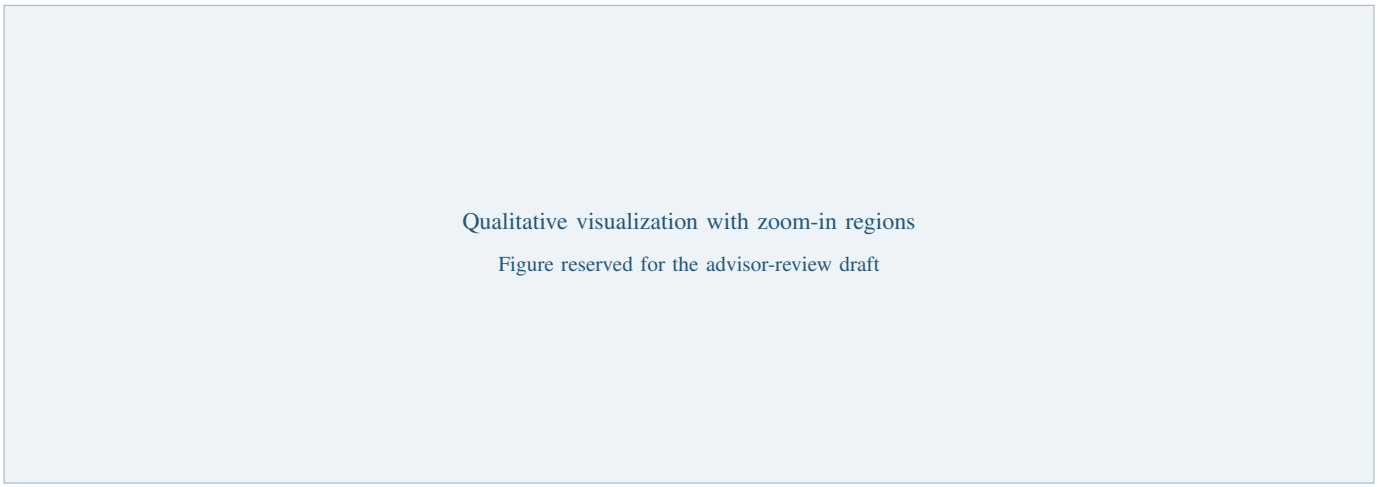


Fig. 6. Visualization. *Pending*

D. Detailed Visualization

Draft note: Detailed visualization in Fig. 6, followed by a zoom-in explanation.

V. Conclusion

This paper presents AS-Depth, a heterogeneous depth-completion system that decouples representation encoding on Thor from depth decoding on FPGA. Its task-level runtime overlaps consecutive frames, while DAQ coordinates low-precision encoder MLP computation, the cross-device representation, and the dedicated decoder datapath. The final manuscript will close by reporting the measured throughput, metric-depth accuracy, and energy efficiency achieved by this combination on HAMMER and ClearPose.

References

[1] J. Uhrig, N. Schneider, L. Schneider, U. Franke, T. Brox, and A. Geiger, "Sparsity invariant CNNs," in *2017 International Conference on 3D Vision (3DV)*, 2017, pp. 11–20.

[2] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2704–2713.

[3] Y. Umuroglu, N. J. Fraser, G. Gambardella, M. Blott, P. Leong, M. Jahre, and K. Vissers, "FINN: A framework for fast, scalable binarized neural network inference," in *Proceedings of the 2017 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays*, 2017, pp. 65–74.